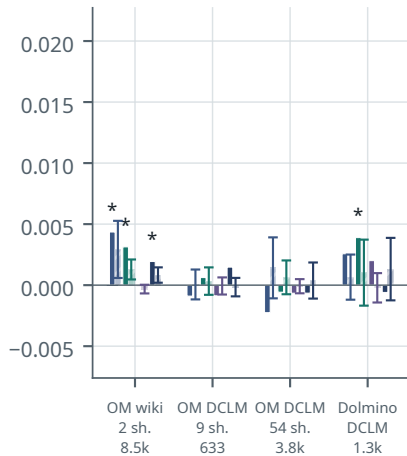
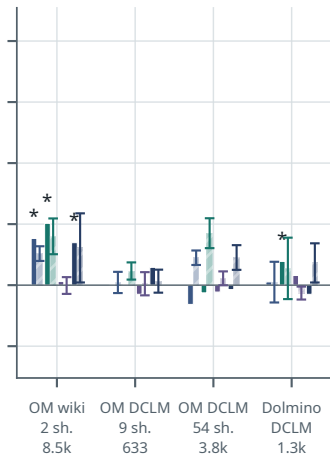


held-out ΔKL (nats per row)

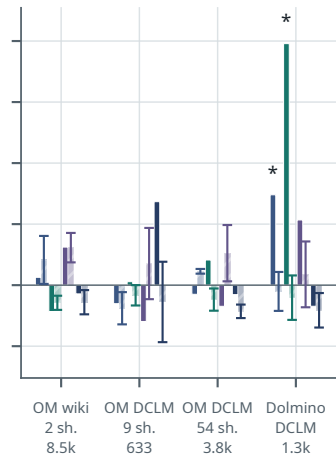
(a) modulation $\times$ document



(b) modulation $\times$ window



(c) harmonic $\times$ window



OLMo-2-1B

Gemma-2-2B

Qwen2.5-3B

OLMo-2-7B

Wikipedia thinned: five-seed mean $\pm$ SD